

Thermodynamic Bounds and Mass-Trade Criteria for Heat Rejection in Orbital Data Centers

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Abstract

Orbital data centers must reject waste heat by far-field thermal radiation, and the resulting radiator area requirements are widely regarded as the binding engineering constraint on the concept. We derive, within a gray-body effective-sink radiator model, a set of exact thermodynamic results governing this constraint. First, we prove that the Carnot efficiency evaluated at the environmental sink temperature is unattainable by any heat engine with a material cold reservoir at finite radiator area and positive throughput, and we characterize both limiting routes (infinite area at fixed throughput; vanishing throughput at fixed area). Second, we show that radiator area per unit work output is minimized, along the reversible lower envelope, at a cold-side temperature $T_c^* = \frac{3}{4}T_h$ with a 25% efficiency ceiling, and we give an exact implicit characterization of the nonzero-sink optimum, $T_c^*/T_h = (3 + q^4)/4$ with $q = T_s^{\text{eff}}/T_c^*$, whose fractional shift above $\frac{3}{4}T_h$ is exactly $q^4/3$. Third, we prove that converting waste heat to work before rejection multiplies required radiator area by at least $(T_h/T_c)^3$, with equality only for reversible conversion at zero sink temperature. Fourth, we show no cyclic system can sustain its compute load solely from its own waste heat. Finally, we state reduced-order mass-trade criteria for radiator-type selection, heat-pump inclusion, and topping-cycle inclusion. All quantitative claims are verified by an accompanying machine-executable test suite (Python and Wolfram Language). The central design implication is that orbital thermal management is a *temperature-architecture* problem: within the model, the temperature at which heat finally leaves the system dominates fixed-temperature efficiency optimizations.

1 Introduction

In June 2026, the World Economic Forum cautioned an industry actively raising capital for orbital data centers that the physics of cooling in orbit “may be more complex than some think” [12]. The caution is quantifiable. A computing facility in vacuum has no air and no water to carry heat away; essentially all waste heat must leave by far-field thermal radiation, governed by the Stefan–Boltzmann law, and at server-class temperatures the required radiating area is of order 10^3 m^2 per megawatt ([11]; Corollary 2 supplies the internal reference calculation of 2630 m^2 per megawatt under stated assumptions). Every square meter of it must be launched.

Discussion of this problem in the technical and trade literature frequently couples two further ideas: that the 2.7 K cosmic microwave background is an enormously favorable cold reservoir for waste-heat-to-electricity conversion, and that such conversion could substantially offset the power and cooling budget of an orbital facility. Both ideas require care. The cold side of any real heat

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engine is a physical radiator whose temperature is set by its own radiative balance rather than by the sky temperature; and extracting work before rejection *lowers* the rejection temperature, which by the quartic law *increases* the radiator area required per watt rejected.

This paper makes those folk observations precise. Working within an explicitly scoped gray-body, isothermal, effective-sink radiator model (Section 2), we prove a hierarchy of results separated into three levels: **Level A** model identities (Section 3), **Level B** reversible thermodynamic lower bounds (Section 4), and **Level C** reduced-order architecture decision criteria with empirical inputs (Section 5). The main results are:

1. **Non-attainability** (Theorem 1): no engine with a material cold reservoir attains the sink-temperature Carnot efficiency at finite radiator area and positive heat or work throughput; the limit is approached only as $A \rightarrow \infty$ at fixed throughput, or as throughput vanishes at fixed area. Efficiencies numerically close to the limit are not forbidden but command extreme area (a worked example at 99% efficiency requires $\sim 6.4 \times 10^9 \text{ m}^2$ per megawatt).
2. **The $\frac{3}{4}$ rule** (Theorem 2): along the reversible lower envelope, radiator area per unit work is minimized at $T_c^* = \frac{3}{4}T_h$, with efficiency ceiling 25%. For irreversible engines with efficiency a fixed fraction of Carnot, the optimum shifts upward (we tabulate examples). This optimum is distinct in both objective and transfer law from the Curzon–Ahlborn maximum-power result [1], and is consistent with radiator-limited space-power optimization studies [4, 5].
3. **Exact nonzero-sink correction** (Theorem 3): the optimum satisfies the quintic $4T_c^5 - 3T_hT_c^4 - T_h(T_s^{\text{eff}})^4 = 0$, equivalently $T_c^*/T_h = (3 + q^4)/4$ with $q = T_s^{\text{eff}}/T_c^*$, giving an exact fractional shift $q^4/3$ above $\frac{3}{4}T_h$ (below 2% for $q \leq 0.49$).
4. **Conversion area penalty** (Corollary 3): any engine rejecting at $T_c < T_h$ requires at least $(T_h/T_c)^3$ times the radiator area of direct rejection at T_h ; the penalty is strictly larger for irreversible engines and for any nonzero sink. At the $\frac{3}{4}$ optimum the minimum penalty is $(4/3)^3 \approx 2.37$ for 25% recovery.
5. **No self-powering** (Theorem 5): a cyclic system cannot sustain a positive compute load solely by reconvertng its own waste heat; recirculated recovery sustains at most $1/(1 - \eta)$ watts of compute per external watt.

Thermoradiative (negative-illumination) devices [2, 3], in which part of the outgoing photon flux is converted directly to electrical work, fall outside the premise of a material cold reservoir in Theorem 1 and require a separate detailed-balance treatment; we delimit but do not analyze them. High-temperature wide-bandgap electronics, which set how far the hot side can be raised in practice, are likewise treated as empirical inputs [9, 10].

All quantitative claims in this paper are verified by a machine-executable test suite provided as supplementary material in two independent forms: a Python assertion script and a Wolfram Language symbolic verification suite. The manuscript was developed through an iterative adversarial workflow across multiple frontier AI systems with computer-algebra verification; see Acknowledgments.

2 Model, Axioms, and Definitions

2.1 Scope

All results are exact within the following model: a gray, diffuse, isothermal radiating surface of hemispherical infrared emissivity $\varepsilon \in (0, 1]$ and total emitting area A (a two-sided panel of

planform area A_p has $A = 2A_p$), exchanging with a lumped radiative environment characterized by a single view-factor-weighted effective sink temperature $T_s^{\text{eff}} = F^{1/4}T_s$, where T_s is an environmental brightness temperature and $F \in [0, 1]$ a view-factor weighting. Where environmental and surface effects can be represented by fixed effective properties and a lumped sink temperature, they alter only model parameters; where absorbed loads, spectral properties, geometry, or temperatures depend on the design variables, they may also alter the functional form of the heat balance and the resulting optimum. Direct application to a flight architecture therefore requires separate environmental heat-load and view-factor modeling [6, 7].

Terminal heat rejection to vacuum occurs in this model by far-field electromagnetic radiation. Open-cycle mass ejection and deliberate export of coherent or nonthermal radiation are outside the model. Near-field radiative transfer and vacuum phonon tunneling operate only across sub-wavelength gaps and require a second material body that must itself radiate; they are internal transport mechanisms, not terminal rejection.

The worked sink value $T_s^{\text{eff}} = 220 \text{ K}$ used in examples is an *illustrative* effective sink temperature for low Earth orbit under the lumped model, not a claim about the temperature of space; published lumped values for Earth-viewing surfaces span roughly 200–260 K (e.g., [6, 7]).

2.2 Axioms

Axiom 1 (Gray-body radiative rejection). Net radiated power is

$$P_{\text{net}} = \varepsilon \sigma A (T_c^4 - (T_s^{\text{eff}})^4), \quad \sigma = 5.670374419 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}. \quad (1)$$

Axiom 2 (Second law). Any cyclic engine operating between reservoirs at $T_h > T_c$ satisfies $\eta \leq 1 - T_c/T_h$; any cyclic heat pump satisfies $\text{COP}_c = Q_c/W \leq T_c/(T_h - T_c)$. Equivalently (Kelvin–Planck), no cyclic device converts heat drawn from a single thermal reservoir entirely into work.

Axiom 3 (First law). For an engine, $Q_h = W + Q_c$; for a heat pump, $Q_h = Q_c + W$.

Axiom 4 (Lumped compute heat model). In steady operation, electrical energy not exported in signals, stored energy, or other retained forms ultimately appears as heat; those fractions are assumed negligible, so compute power P dissipates as heat at the source temperature. Net rejection requires $T_c > T_s^{\text{eff}}$.

Definition 1 (Reversible engine). An engine attaining equality in the Carnot bound, $\eta = 1 - T_c/T_h$. With Axiom 3 this yields the reversible heat split $Q_c/Q_h = T_c/T_h$. All uses of this equality below are labeled reversible; generic engines receive the corresponding inequality bound.

Level A results follow from Axioms 1 and 3; Level B results from Axioms 1–4 plus Definition 1 where labeled; Level C results additionally require the reduced-order mass models and empirical parameters identified in Section 5.

3 Level A: Model Identities

Lemma 1 (Radiator area requirement). *Rejecting heat flow $Q_c > 0$ at radiator temperature T_c requires*

$$A = \frac{Q_c}{\varepsilon \sigma (T_c^4 - (T_s^{\text{eff}})^4)}. \quad (2)$$

Proof. Invert (1) with $P_{\text{net}} = Q_c$; positive and well defined by Axiom 4. $\square$

Corollary 1 (Area ratio between rejection temperatures). *For equal heat load and emissivity, rejecting at T_2 instead of T_1 (with $T_2 > T_1 > T_s^{\text{eff}}$) changes area by the exact factor*

$$R = \frac{A_1}{A_2} = \frac{T_2^4 - (T_s^{\text{eff}})^4}{T_1^4 - (T_s^{\text{eff}})^4}, \quad (3)$$

and the zero-sink approximation $R_0 = (T_2/T_1)^4$ carries exact relative deviation

$$\frac{R}{R_0} - 1 = \frac{(T_s^{\text{eff}}/T_1)^4 - (T_s^{\text{eff}}/T_2)^4}{1 - (T_s^{\text{eff}}/T_1)^4} < \frac{(T_s^{\text{eff}}/T_1)^4}{1 - (T_s^{\text{eff}}/T_1)^4}. \quad (4)$$

Worked example ($T_1 = 293 \text{ K}$, $T_2 = 600 \text{ K}$, $T_s^{\text{eff}} = 220 \text{ K}$): $R_0 = 17.585$ while $R = 25.312$ exactly ($R = 6697760000/264604779$); the exact ratio is 43.9% greater than the zero-sink estimate, equivalently the estimate is 30.5% below the exact value. The warm sink penalizes the colder radiator disproportionately, so hot rejection is more advantageous than the idealized quartic ratio suggests.

Corollary 2 (Reference scale). $Q_c = 1 \text{ MW}$ at $T_c = 293 \text{ K}$, $\varepsilon = 0.91$, $T_s^{\text{eff}} \approx 0$ requires $A = 2630 \text{ m}^2$ of emitting surface, i.e. $\sim 1315 \text{ m}^2$ of two-sided panel planform, consistent with the $\sim 1200 \text{ m}^2/\text{MW}$ figures quoted in the trade literature for slightly different assumptions [11].

Lemma 2 (COP identity). *For any heat pump, $\text{COP}_h = \text{COP}_c + 1$.*

Proof. Divide $Q_h = Q_c + W$ (Axiom 3) by W . □

4 Level B: Reversible Thermodynamic Lower Bounds

4.1 Non-attainability of sink-temperature Carnot efficiency

Theorem 1 (Non-attainability). *For any finite area A and strictly positive rejected heat flow Q_c , Lemma 1 forces*

$$T_c^4 = (T_s^{\text{eff}})^4 + \frac{Q_c}{\varepsilon \sigma A} > (T_s^{\text{eff}})^4, \quad \text{hence} \quad \eta \leq 1 - \frac{T_c}{T_h} < 1 - \frac{T_s^{\text{eff}}}{T_h}. \quad (5)$$

The sink-temperature Carnot value is not attained by any system with finite area and positive heat or work throughput. It is approached only along two limiting routes: (i) at fixed nonzero heat or work throughput, as $A \rightarrow \infty$; (ii) at fixed finite area, as throughput $Q_c \rightarrow 0$.

Proof. The strict inequality is immediate from Lemma 1. *Route (i), fixed heat throughput:* at fixed $Q_c > 0$, $T_c^4 - (T_s^{\text{eff}})^4 = Q_c/(\varepsilon \sigma A) \rightarrow 0$ iff $A \rightarrow \infty$. *Route (i), fixed work throughput:* for fixed $W > 0$, Axiom 2 gives $Q_c/W \geq T_c/(T_h - T_c)$, hence

$$\frac{A}{W} \geq \frac{T_c}{\varepsilon \sigma (T_h - T_c) (T_c^4 - (T_s^{\text{eff}})^4)}, \quad (6)$$

which diverges as $T_c \rightarrow (T_s^{\text{eff}})^+$ when $T_s^{\text{eff}} > 0$; for $T_s^{\text{eff}} = 0$ the bound reduces to $1/[\varepsilon \sigma (T_h - T_c) T_c^3]$, which diverges as $T_c \rightarrow 0^+$. *Route (ii):* at fixed $A < \infty$, $T_c^4 - (T_s^{\text{eff}})^4 \rightarrow 0$ iff $Q_c \rightarrow 0$. □

Remark 1 (Near-limit efficiencies are extreme-area, not impossible). Claims that obtain $\sim 99\%$ efficiency merely by substituting the 2.7 K cosmic background for the material cold-reservoir temperature are incomplete rather than impossible: a finite-throughput design must solve for $T_c > T_s$ and the associated area. For $T_h = 300 \text{ K}$, $T_s = 2.7 \text{ K}$, $T_c = 3.0 \text{ K}$, reversible efficiency is 99.0%, and at $W = 1 \text{ MW}$ the rejected heat is $Q_c = WT_c/(T_h - T_c) \approx 10.1 \text{ kW}$, requiring ideal blackbody area $A \approx 6.4 \times 10^9 \text{ m}^2$ (finite, but roughly six million times the reference scale of Corollary 2, and far outside the design space this model contemplates).

Remark 2 (Direct radiative converters). Thermoradiative and negative-illumination devices [2, 3], in which part of the outgoing photon flux is converted directly to electrical work, fall outside the premise of a material cold reservoir in Theorem 1 and require a separate detailed-balance treatment. No claim of uniqueness is made for that architecture, and no second-law exception is implied.

4.2 The $\frac{3}{4}$ rule

Theorem 2 (Reversible lower-envelope optimum, zero sink). *For any engine producing work W and rejecting at T_c through a radiator with $T_s^{\text{eff}} = 0$,*

$$\frac{A}{W} = \frac{1 - \eta}{\eta \varepsilon \sigma T_c^4} \geq \frac{1}{\varepsilon \sigma (T_h T_c^3 - T_c^4)}, \quad (7)$$

with equality iff the engine is reversible. The right-hand side is minimized at

$$T_c^* = \frac{3}{4}T_h, \quad \eta(T_c^*) = 25\%. \quad (8)$$

This is the optimum of the reversible lower envelope; it is not the operating optimum of every irreversible engine.

Proof. Axiom 3 gives $Q_c = W(1 - \eta)/\eta$, and Axiom 2 gives $(1 - \eta)/\eta \geq T_c/(T_h - T_c)$, establishing (7). Minimizing the bound is equivalent to maximizing $f(T_c) = T_h T_c^3 - T_c^4$ on $(0, T_h)$:

$$f'(T_c) = T_c^2(3T_h - 4T_c) = 0 \Rightarrow T_c^* = \frac{3}{4}T_h, \quad f''(T_c^*) = -\frac{9}{4}T_h^2 < 0,$$

and $f \rightarrow 0$ at both endpoints, so the interior critical point is the global maximum of f , hence the global minimum of the bound. The reversible efficiency there is $1 - \frac{3}{4} = 25\%$. $\square$

Remark 3 (Irreversible engines). For an engine with efficiency law $\eta = a(1 - T_c/T_h)$, $0 < a < 1$, the area-per-work optimum shifts upward; representative values of the optimal $y^* = T_c/T_h$ are:

a (fraction of Carnot)	optimal T_c/T_h
1.0	0.7500
0.8	0.7645
0.5	0.7808

Engine selection in practice requires the engine's actual $\eta(T_c)$ law.

Remark 4 (Relation to Curzon–Ahlborn). The Curzon–Ahlborn efficiency $1 - \sqrt{T_c/T_h}$ [1] arises in an endoreversible maximum-power model with finite-rate linear heat transfer. Theorem 2 optimizes a different objective (radiator area per unit work) under a radiative T^4 rejection law; no agreement should be expected. Optima in the neighborhood of $\frac{3}{4}T_h$ are reproduced by radiator-limited space-power studies [4, 5].

Corollary 3 (Conversion area penalty). *Relative to direct rejection of Q_h at T_h , any engine rejecting at $T_c < T_h$ requires, at $T_s^{\text{eff}} = 0$,*

$$\frac{A_{\text{engine}}}{A_{\text{direct}}} = (1 - \eta) \left(\frac{T_h}{T_c} \right)^4 \geq \left(\frac{T_h}{T_c} \right)^3, \quad (9)$$

with equality only in the reversible limit. At $T_c = \frac{3}{4}T_h$ the minimum penalty is $(4/3)^3 \approx 2.370$ for 25% recovery; irreversibility increases it (e.g. $a = 0.8$ at the same T_c gives ≈ 2.528). With a common nonzero sink the exact expression is

$$\frac{A_{\text{engine}}}{A_{\text{direct}}} = (1 - \eta) \frac{T_h^4 - (T_s^{\text{eff}})^4}{T_c^4 - (T_s^{\text{eff}})^4} \geq \left(\frac{T_h}{T_c}\right)^3 \frac{1 - (T_s^{\text{eff}}/T_h)^4}{1 - (T_s^{\text{eff}}/T_c)^4} > \left(\frac{T_h}{T_c}\right)^3, \quad (10)$$

so the cubic penalty is a universal lower bound whose equality requires both reversibility and $T_s^{\text{eff}} = 0$.

Proof. $A_{\text{direct}} = Q_h/[\varepsilon\sigma(T_h^4 - (T_s^{\text{eff}})^4)]$ and $A_{\text{engine}} = Q_h(1 - \eta)/[\varepsilon\sigma(T_c^4 - (T_s^{\text{eff}})^4)]$ by Lemma 1 and Axiom 3; apply $1 - \eta \geq T_c/T_h$ (Axiom 2). For $T_s^{\text{eff}} > 0$, $T_c < T_h$ makes the final fraction exceed unity. $\square$

4.3 Exact nonzero-sink optimum

Theorem 3 (Implicit characterization of the nonzero-sink optimum). *With effective sink $T_s^{\text{eff}} > 0$, the reversible-envelope optimum T_c^* satisfies the quintic stationarity equation*

$$4T_c^5 - 3T_hT_c^4 - T_h(T_s^{\text{eff}})^4 = 0, \quad (11)$$

equivalently the exact implicit characterization

$$\frac{T_c^*}{T_h} = \frac{3 + q^4}{4}, \quad q \equiv \frac{T_s^{\text{eff}}}{T_c^*}, \quad (12)$$

with exact fractional shift above $\frac{3}{4}T_h$ equal to $q^4/3$. The optimum is unique in (T_s^{eff}, T_h) and strictly increasing in T_s^{eff} ; the shift is at most $0.49^4/3 \approx 1.9216\%$ for $q \leq 0.49$.

Proof. By Lemma 1 and Definition 1, minimizing A/W is equivalent to maximizing

$$g(T_c) = \frac{(T_h - T_c)(T_c^4 - (T_s^{\text{eff}})^4)}{T_c} = T_hT_c^3 - T_c^4 - \frac{T_h(T_s^{\text{eff}})^4}{T_c} + (T_s^{\text{eff}})^4.$$

Then $g'(T_c) = 3T_hT_c^2 - 4T_c^3 + T_h(T_s^{\text{eff}})^4/T_c^2$; setting $g' = 0$ and multiplying by $T_c^2 > 0$ yields (11). Dividing (11) by $4T_hT_c^4$ gives (12), and the fractional shift is $(T_c^* - \frac{3}{4}T_h)/(\frac{3}{4}T_h) = (q^4/4)/(3/4) = q^4/3$, monotone in q . In reduced variables $y = T_c/T_h$, $r = T_s^{\text{eff}}/T_h$, stationarity reads $4y^5 - 3y^4 = r^4$; for $y \geq \frac{3}{4}$, $\frac{d}{dy}(4y^5 - 3y^4) = 4y^3(5y - 3) > 0$, so the physical root is unique and strictly increasing in T_s^{eff} . Since $g(T_s^{\text{eff}}) = g(T_h) = 0$ with $g > 0$ between, the stationary point is the interior maximum. $\square$

Remark 5 (Computation). The fixed-point map $\Phi(T) = \frac{T_h}{4}[3 + (T_s^{\text{eff}}/T)^4]$ has $|\Phi'(T_c^*)| = 4q^4/(3 + q^4) < 1$ for all physical $q < 1$, so iteration of (12) is locally convergent. At the illustrative design point $T_h = 600$ K, $T_s^{\text{eff}} = 220$ K: $T_c^* = 457.98675408138325$ K (high-precision evaluation of the exact algebraic root; cross-checked by Newton iteration with quintic residual below 10^{-25} and by fixed-point iteration), a shift of $+1.7748\%$ matching $q^4/3$ to all computed digits. The $\frac{3}{4}$ rule therefore survives at sub-2% accuracy in the relevant regime, with an analytic error formula in place of numerical tables. No claim is made regarding solvability of (11) by radicals.

4.4 Heat pumps and self-powering

Theorem 4 (Heat-pump overhead and area ratio). *Pumping the thermal load Q_c extracted at T_c up to rejection at T_h requires work $W/Q_c = 1/\text{COP}_c \geq (T_h - T_c)/T_c$, and the radiator must reject $Q_h = Q_c(1 + 1/\text{COP}_c)$. Pumping from unpumped rejection temperature T_1 to pumped temperature T_2 with common sink T_s^{eff} , equal emissivity and view geometry,*

$$\frac{A_{\text{pump}}}{A_{\text{direct}}} = \left(1 + \frac{1}{\text{COP}_c}\right) \frac{T_1^4 - (T_s^{\text{eff}})^4}{T_2^4 - (T_s^{\text{eff}})^4}. \quad (13)$$

Worked example ($T_1 = 353 \text{ K}$, $T_2 = 520 \text{ K}$, $T_s^{\text{eff}} = 220 \text{ K}$, $\text{COP}_c = 1.15$): exact ratio 0.348 (zero-sink approximation 0.397). At $T_c = 353 \text{ K}$, $T_h = 520 \text{ K}$ the Carnot bounds are $\text{COP}_c \leq 353/167 \approx 2.114$ and $\text{COP}_h \leq 520/167 \approx 3.114$, consistent with Lemma 2; minimum overhead is 0.473 W per watt of load. The range $\text{COP}_c \approx 1.0\text{--}1.3$ used in examples is an empirical parameter for vapor-compression machines at $\sim 170 \text{ K}$ lift, not derived from the axioms.

Proof. Axioms 2–3, Lemmas 1 and 2. □

Theorem 5 (No waste-heat self-powering). *No cyclic system can sustain a positive compute load solely by converting its own internally generated waste heat back into work. Under Axiom 4 with compute power P , recovery efficiency $\eta < 1$ (strict by Theorem 1 at finite area and positive throughput), and recovered work recirculated to the bus,*

$$P_{\text{ext}} = P(1 - \eta) > 0. \quad (14)$$

External energy resources other than the waste heat itself (solar, nuclear, beamed, or stored chemical) are outside this claim.

Proof. Steady-state bus balance: $P = P_{\text{ext}} + W_{\text{rec}}$ with $W_{\text{rec}} = \eta P$, so $P_{\text{ext}} = P(1 - \eta) > 0$. Self-powering would require $\eta \geq 1$: a cyclic device converting heat from a single effective reservoir entirely into work, contradicting the Kelvin–Planck statement (Axiom 2). □

Remark 6. Recirculation sustains $P/P_{\text{ext}} = 1/(1 - \eta)$ watts of compute per external watt ($\eta = 0.25$ gives 1.33). Recovery therefore reduces the external supply requirement by a bounded factor and cannot eliminate it.

5 Level C: Architecture Decision Criteria

The following are decision criteria under explicitly reduced mass models with empirical inputs; their verdicts flip with parameter values, while their algebraic validity is parameter-independent. Empirical parameters appearing below (areal densities ρ_A , specific masses, COP values, margin factors δ) are catalogued in the supplementary material and must be normalized to equal delivered duty.

Proposition 1 (Radiator selection). *Let ρ_A denote effective system areal density normalized to equal delivered rejection capacity (equal rejected power, operating temperature, effective emissivity, view geometry, lifetime and availability). Droplet radiators are mass-favorable over solid panels iff*

$$\rho_{A,\text{droplet}} (1 + \delta_{\text{capture}}) < \rho_{A,\text{solid}} (1 + \delta_{\text{debris}}). \quad (15)$$

Proposition 2 (Heat-pump inclusion: necessary first-order condition). *Under the stated reduced mass model, including pump and ancillary masses, active thermal upgrade is mass-favorable only if*

$$\rho_A (A_{\text{direct}} - A_{\text{pump}}) > \frac{Q_c}{\text{COP}_c} m_{\text{array}} + M_{\text{pump}} + M_{\text{ancillary}}, \quad (16)$$

with A_{pump} from Theorem 4 and m_{array} the array specific mass (kg per electrical watt). A full verdict requires the complete mass ledger.

Proposition 3 (Topping-cycle inclusion). *With engine specific mass per watt of thermal input $s_h = M_{\text{engine}}/Q_h$, the engine is mass-favorable under the reduced model iff*

$$s_h < \eta m_{\text{array}} + \rho_A \frac{A_{\text{direct}} - A_{\text{engine}}}{Q_h}, \quad (17)$$

equivalently, per watt recovered ($s_e = s_h/\eta$), $s_e < m_{\text{array}} + \rho_A (A_{\text{direct}} - A_{\text{engine}})/(\eta Q_h)$. Since $A_{\text{engine}} > A_{\text{direct}}$ (Corollary 3), the area term is negative: the engine must overcome both its own mass and its area penalty, with $\eta \leq \frac{1}{4}$ at the reversible-envelope optimum.

6 Discussion

Within the adopted radiator model, increasing rejection temperature reduces required emitting area by the exact factor of Corollary 1, and a warm sink amplifies the effect. Whether that reduction yields a lower-mass, lower-power, more reliable, or longer-lived complete architecture is governed by the Level C criteria and their empirical inputs, notably the maximum qualified junction temperature of the compute technology: silicon’s practical limit is typically near 100–150 °C, while SiC and GaN devices have demonstrated operation at 500–800 °C [9, 10], making wide-bandgap electronics the natural lever for hot-side elevation. Within the model, rejection temperature is the budget from which every other thermal choice is purchased.

The results bound, rather than recommend, waste-heat recovery: Corollary 3 prices recovery in area (a minimum of $2.37\times$ at 25% recovery), Theorem 5 caps what it can return, and Proposition 3 states the exact condition under which its mass trade closes. Thermoradiative conversion, in which the radiator surface is itself the generator, is the one architecture outside these bounds and merits the separate detailed-balance analysis indicated in Section 4.

Limitations. All results are exact only within the gray-body, isothermal, effective-sink model. Design-dependent absorbed loads, spectral and directional surface properties, non-isothermal radiators, multi-surface view factors, and transient orbital environments can alter the functional form of the heat balance and hence the optima. Flight application requires environment-resolved thermal modeling [6]. The Level C criteria are first-order mass conditions; a complete system trade requires the full mass and reliability ledger.

7 Conclusion

The complexity the World Economic Forum warned of in June 2026 is real, and within the adopted model it is also exactly characterizable. The temperature at which heat finally leaves the system enters the area requirement quartically and dominates fixed-temperature efficiency optimization. The reversible-envelope optimum for combined rejection and recovery sits at $T_c^* = \frac{3}{4}T_h$, with a sink correction of exactly $q^4/3$. Conversion before rejection always costs area. No architecture bootstraps from its own waste heat. These bounds frame the empirical engineering trades that will determine whether orbital computing closes as a system. The bounds are proved and the trades are stated. What remains is engineering.

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Data and Code Availability

Supplementary material accompanying this preprint includes: (i) a Python assertion suite verifying every central numerical claim; (ii) a Wolfram Language symbolic verification suite (stationarity, second-order conditions, limits, exact rationals, and high-precision roots); and (iii) the audited source document with full revision history. The complete package is archived at Zenodo ([doi:10.5281/zenodo.20650894](https://doi.org/10.5281/zenodo.20650894)) and mirrored at <https://github.com/dan-lee-odinson/orbital-thermal-bounds>.

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